

Cisco Packet Tracer / CCNA Lab Series

Chapter 6 of the CCNA study guide - Cisco IOS

- **Goals:**

In this lab series the goal was to familiarize myself with Cisco's IOS. The labs focused on reading outputs of the show interface command, showing the running and start configuration and how to copy running to start. Switch and router configuration such as setting passwords, ssh, enabling secrets, adding messages and reading DCE and DTE. I created this series using Claude AI to build a 10 lab series that progressively increased in difficulty with the last lab being a culmination of the previous 9.

- **Set up:**

The labs comprised of a mix of a 2911 router and a switch, the routers were connected using DCE/DTE cables and the switch connected to a router through Gigabit Ethernet.

- **Lab Notes:**

The lab series was broken down into two sections, labs 1-5 being introductory. They covered the basic requirements for the chapter, entering commands such as "show run" to see the running configuration, going into global configuration to change the hostname of the router, copying that to that starting configuration using "copy run start", interfacing into both console line and VTY lines to set the passwords and enable login, and also setting enable secret for the passwords as well as encrypting passwords using the "service-encrypt password" command.

In labs 6-10 they were a mix of these commands and in addition setting banners to appear before logging in using "MOTD banner #", setting the Domain name, RSA keys and SSH to version 2. I was also tasked with creating a local user account and setting up VTY transportation to allow me to remote in to one router from the other after configuring the DCE/DTE cables, ensuring the clock rate and IP's were properly assigned.

Through this series I learned how to comfortably navigate through the IOS to set up routers and switches, create logins to access both and to remote into routers. I've also learned how to save configurations and erase and reload previous configurations.